This is an untimed collection of practice problems for the cumulative final. It is not intended to match the exam's length, point distribution, or order of questions. Work through the problems before consulting the solutions. Show calculations and explain the reasoning behind conceptual answers. More than one approach may be valid when its assumptions are clearly stated and justified.

Formulas

Search and planning

$$f(n) = g(n) + h(n), \quad h(n) \le h^*(n), \quad h(n) \le c(n, n') + h(n'), \quad h(G) = 0$$

MDPs and RL

$$(\mathcal{S}, \mathcal{A}, T, R, \gamma), \quad T(s' \mid s, a) = P(s' \mid s, a), \quad \sum_{s'} T(s' \mid s, a) = 1, \quad 0 \le \gamma < 1$$

$$Q^*(s,a) = \sum_{s'} T(s' \mid s, a)[R(s, a, s') + \gamma \max_{a'} Q^*(s', a')]$$

$$V^*(s) = \max_a Q^*(s, a), \qquad \pi^*(s) \in \arg\max_a Q^*(s, a)$$

$$V_{k+1}(s) = \max_a \sum_{s'} T(s' \mid s, a)[R(s, a, s') + \gamma V_k(s')]$$

$$V^{\pi_k}(s) = \sum_{s'} T(s' \mid s, \pi_k(s))[R(s, \pi_k(s), s') + \gamma V^{\pi_k}(s')]$$

$$\pi_{k+1}(s) \in \arg\max_a \sum_{s'} T(s' \mid s, a)[R(s, a, s') + \gamma V^{\pi_k}(s')]$$

$$V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$$

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Optimization

$$x_{k+1} = x_k - \alpha f'(x_k), \qquad f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \tfrac{1}{2} f''(x_0)(x - x_0)^2$$

$$F(x) = \frac{1}{N} \sum_{i=1}^{N} f_i(x), \quad F'(x) = \frac{1}{N} \sum_{i=1}^{N} f_i'(x), \quad \widehat{g}(x) = \frac{1}{m} \sum_{j=1}^{m} f_{i_j}'(x) \approx F'(x)$$

Transforms

$${}^a\mathbf{T}_b = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} p_a \\ 1 \end{bmatrix} = {}^a\mathbf{T}_b \begin{bmatrix} p_b \\ 1 \end{bmatrix}, \quad {}^a\mathbf{T}_c = {}^a\mathbf{T}_b {}^b\mathbf{T}_c, \quad R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculus

$$(x^n)' = nx^{n-1}, \quad (\sin x)' = \cos x, \quad (\cos x)' = -\sin x$$

$$\sin 0 = 0, \quad \cos 0 = 1, \quad \sin(\pi/2) = 1, \quad \cos(\pi/2) = 0$$

Beliefs and filtering

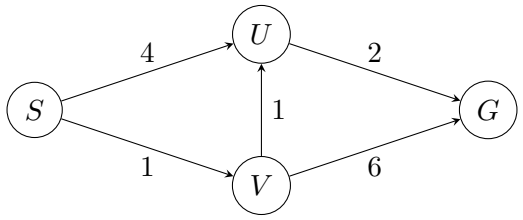
$$B_t(x_i) = P(X_t = x_i \mid z_{0:t}, u_{1:t}), \qquad P(x_i \mid z) = \frac{P(z \mid x_i) P(x_i)}{\sum_j P(z \mid x_j) P(x_j)}$$

$$\overline{B}_t(x_i) = \sum_j T(x_i \mid x_j, u_t) B_{t-1}(x_j)$$

$$\widetilde{B}_t(x_i) = P(z_t \mid x_i) \overline{B}_t(x_i), \qquad B_t(x_i) = \frac{\widetilde{B}_t(x_i)}{\sum_j \widetilde{B}_t(x_j)}$$

Question 1: Following and modifying a heuristic

The directed graph has edges $S \to U$ of cost 4, $S \to V$ of cost 1, $V \to U$ of cost 1, $U \to G$ of cost 2, and $V \to G$ of cost 6. There are no other edges. The heuristic at (S, U, V, G) is $(4, 2, 3, 0)$. A* selects the smallest $g + h$, breaking ties by smaller h, then alphabetical label. Reopen a node when its path cost improves, and stop when the goal is selected.



(a) Trace A* through goal selection. Give the frontier priorities after each nonterminal expansion and the returned path.

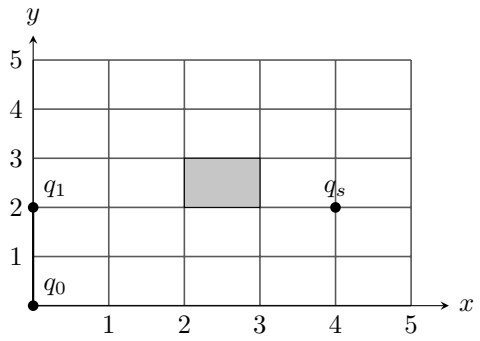
Answer / work

(b) Change only $h(V)$ to 2. Determine admissibility and consistency. If they differ, explain how that is possible.

Answer / work

Question 2: Steering a robot with a footprint

A rectangular robot translates without rotating. Its width is 1 along x and its height is 2 along y. Its configuration is its center. An obstacle occupies $[2,3] \times [2,3]$. Contact counts as collision. An RRT has nodes $q_0 = (0,0)$ and $q_1 = (0,2)$ and samples $q_s = (4,2)$. Use Euclidean nearest-node selection and straight-line steering.



(a) Draw and state the set of center positions in collision. Which node is nearest to the sample?

Answer / work

(b) Starting independently from the original tree, test maximum extension lengths 1 and 4. Give each endpoint and whether the extension is accepted. Explain why endpoint testing is insufficient.

Answer / work

Question 3: Two ways to improve a policy

An MDP has nonterminal states A, B and terminal state Z. At A, stop gives reward 2 and terminates; continue gives reward 0 and enters B. At B, finish gives reward 6 and terminates. Use $\gamma = 1/2$ and terminal value zero.

(a) Starting with $V_0 = 0$, perform two synchronous VI sweeps. Show the candidate action values at A in each sweep.

Answer / work

(b) Start PI from the policy that stops at A. Evaluate it exactly, then improve it. Is the improved policy already optimal here? Explain.

Answer / work

Question 4: A model and two estimates from the same episodes

A fixed policy produced two complete episodes,

$$A \xrightarrow{1} B \xrightarrow{2} Z, \qquad A \xrightarrow{3} Z.$$

State Z is terminal. Use $\gamma = 1/2$.

(a) Estimate transition probabilities for this policy by counting observed transitions. Give the reward for each observed transition type, then solve for the values in this empirical model.

Answer / work

(b) Compute direct-evaluation estimates by averaging observed returns. Then process the episodes in their given order with TD, starting at zero and using learning rate 1/2. Show each update.

Answer / work

(c) Explain why the results do not establish that the policy is optimal.

Answer / work

Question 5: Learning action values from a short log

Initialize every Q-entry to zero. Use $\alpha = \gamma = 1/2$ and process these records in order. Z is terminal.

State	Action	Reward	Next state
S	left	2	A
A	finish	4	Z
S	left	0	A

(a) Compute the Q-learning sample and new table entry for each record. All unlisted entries remain zero.

Answer / work

(b) An untried action right at S still has value zero. Does the greedy choice prove left is better? Would a larger learning rate supply the missing evidence?

Answer / work

Question 6: A local model and sampled gradient steps

Let $f(x) = e^x + x^2 - 3x$. Construct its quadratic Taylor model $m(x)$ about $x_0 = 0$. Use this same model for all updates below.

(a) Compute $f(0)$, $f'(0)$, and $f''(0)$, then write $m(x)$.

Answer / work

(b) Starting at $x = 1$, take one gradient-descent step on m with learning rate $\alpha = 1/2$. Compute the model's value before and after the step. Use m', not f', for this update.

Answer / work

(c) Instead of using $m'(1)$, suppose a sampled update uses one of the supplied gradient estimates $g_1 = -1$ and $g_2 = 3$, each with probability $1/2$. Starting separately from $x = 1$ in each case, compute the two updates $x_{\text{sample}} = 1 - \alpha g_i$ using the same $\alpha = 1/2$. Compare their average with part (b). Does either individual update have to equal the full-gradient update?

Answer / work

Question 7: Composing frames along an arm

A base frame b is rotated 90° about world z and translated by $(1, 2, 0)$ in the world. A link frame ℓ has the same orientation as the base and origin $(2, 0, 0)$ in the base frame. Its tip has link-frame coordinates $(1, 0, 0)$.

(a) Write both 4×4 transforms, compose them, and compute the tip's world coordinates. Label the source and destination frames.

Answer / work

(b) Why is adding $(1, 2, 0) + (2, 0, 0) + (1, 0, 0)$ incorrect?

Answer / work

Question 8: Choosing methods from available information

Consider A*, PRM, RRT, VI, PI, fixed-policy V-learning, Q-learning, and trajectory optimization. Several proposals may be defensible. For each setting, recommend an approach, explain why the available information supports it, and identify one limitation. Two or three sentences per setting are sufficient. You may combine methods.

(a) A robot receives many start-goal requests in a known static building. Collision checking is available and preprocessing is allowed. *Hint: Consider what could be built once and reused, and how a route would be found within that representation.*

Answer / work

(b) Actions have uncertain outcomes. A small finite transition and reward model is known, and a policy is needed for every state. Later, only recorded episodes under one fixed policy are available. Recommend a method for the known-model setting and a method for estimating the recorded policy's value. *Hint: Distinguish improving a policy from evaluating one. Which methods need a transition model, and which can use observed experience?*

Answer / work

(c) A geometric route is known, but executing it produces abrupt motions and violates acceleration limits. A dynamics model is available. *Hint: The route is already known. Consider a method that can adjust its timing and motion while accounting for actuator constraints.*

Answer / work

Question 9: Filtering in a long aisle

A robot drives through a long straight aisle with identical parallel walls. Its lidar sees the side walls but neither end of the aisle and no distinctive features along its length. The robot uses wheel odometry and lidar in an EKF with nonlinear motion and measurement models. Its lateral position is well localized, but uncertainty about its progress along the aisle grows.

(a) Explain how the filter uses a motion command to predict the pose belief, then uses a lidar reading to correct it. Why should a difference between predicted and measured range not be treated entirely as pose error?

Answer / work

(b) Explain why the scan can constrain the lateral position but not progress along the aisle. Would replacing the EKF with a UKF, or adding more particles to a particle filter, necessarily resolve this uncertainty? Give one measurement that would provide the missing information.

Answer / work

Question 9 (continued)

(c) ★ (required for 281; optional for 81) Suppose the pose belief is concentrated near a good estimate. Compare how an EKF and a UKF propagate it through a nonlinear model, giving one benefit and one limitation of each. Does Gaussian noise make either method exact for every nonlinear model? No equations or derivations are required.

Answer / work

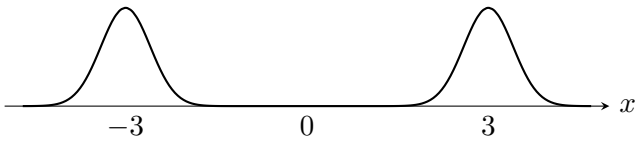
Question 10: Planning Scenarios

Sketch a two-dimensional environment in which A* on a sufficiently fine grid could find a route more reliably within a fixed computation budget than a basic RRT with uniform sampling and 5% goal bias. Identify the start, goal, and obstacles. Explain what makes the environment difficult for RRT, then describe a change that could reduce that difficulty. State any assumptions needed for the grid comparison.

Answer / work

Question 11: Two modes and their average

A robot is equally likely to be near position $x = -3$ or $x = 3$ along a corridor. Each cluster is narrow, and the region near $x = 0$ is a wall. The sketch shows the qualitative posterior density.



(a) What does the posterior mean report? Explain why reporting only that point can be misleading. Can a single Gaussian faithfully represent the two narrow peaks?

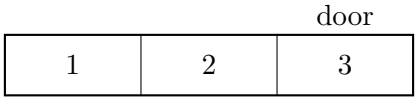
Answer / work

(b) Compare a standard single-Gaussian UKF with a particle filter for this belief. What can cause a particle filter to lose an important hypothesis, and what does the UKF's use of sigma points not imply?

Answer / work

Question 12: Motion, evidence, and a normalized belief

A robot occupies one of three cells. A door is present only in cell 3. Model its hidden cell as an HMM. The next cell depends only on the current cell and motion command, and the reading depends only on the cell occupied when it is taken. Its current belief is $(B(x_1), B(x_2), B(x_3)) = (1/2, 1/4, 1/4)$, where x_i denotes occupancy of cell i.



The robot first attempts to move right. From cells 1 and 2 it moves one cell right with probability $1/2$ and stays with probability $1/2$. From cell 3 it stays with probability 1. It then receives a `door` reading, whose likelihood is $1/4$ in cells 1 and 2, and $3/4$ in cell 3.

(a) Apply only the motion update. Fill every cell of the first row below with its predicted probability $\overline{B}(x_i)$. Do not use the `door` reading in this part. Show which previous cells contribute to each entry.

Answer / work

Cell 1	Cell 2	Cell 3

(b) Multiply the predicted probability in each cell by that cell's likelihood for the observed reading. Fill the row with these unnormalized weights $\widetilde{B}(x_i)$ and compute their sum.

Answer / work

Cell 1	Cell 2	Cell 3

Question 12 (continued)

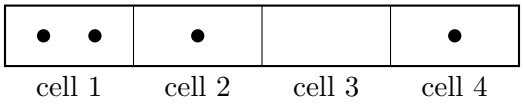
(c) Normalize the three weights to obtain $B_{\text{new}}(x_i)$ and fill every cell below. Leave the results as fractions. Explain why the posterior is not obtained by normalizing the three sensor likelihoods alone.

Answer / work

Cell 1	Cell 2	Cell 3

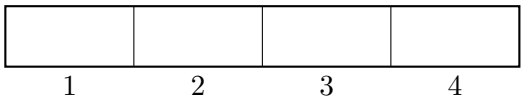
Question 13: Particles and probability in each cell

Four equally weighted particles occupy cells $(1, 1, 2, 4)$ of a four-cell row. A motion rule moves each particle one cell to the right, except that a particle in cell 4 moves one cell left. For this exercise the motion is deterministic. After the move, the likelihood of the observed reading in cells $(1, 2, 3, 4)$ is $(1/2, 1/4, 3/4, 1/2)$.



(a) Draw the particles after motion in the empty row. How much predicted probability mass is represented in each cell?

Answer / work



(b) Multiply each particle's weight by its new cell's likelihood. Then sum the weights within each cell. Enter the unnormalized *cell masses*, not the weight of one particle, in the row below.

Answer / work

Cell 1	Cell 2	Cell 3	Cell 4

Question 13 (continued)

(c) Normalize the cell masses and write them as fractions in the row. Next, draw four particles independently with replacement from the current particles, with selection probabilities given by their normalized weights. Each new particle must copy an existing location. What is the expected number in each cell? Must every set of draws produce those counts?

Answer / work

Cell 1	Cell 2	Cell 3	Cell 4

Question 14: Channels, pooling, and a fully connected layer

An RGB image has shape $6 \times 6 \times 3$. Its three channels contain red, green, and blue values. A convolutional layer applies four learned filters with spatial size 3×3, stride 1, and no padding. Stride 1 moves each filter one pixel at a time. Count only positions where the entire filter fits inside the image. Each filter combines information from all three input channels and produces one output channel. These learned channels need not represent colors. A max-pooling layer next replaces each nonoverlapping 2×2 block in each channel with its largest entry.

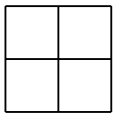
(a) Determine the output dimensions after convolution and after pooling. Explain which operation changes the number of channels and which reduces the spatial dimensions without combining channels.

Answer / work

(b) Pool this 4×4 response from one channel. Fill the 2×2 output grid. Give one piece of information the output loses.

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 4 & 2 & 1 & 5 \\ 0 & 6 & 3 & 2 \\ 1 & 2 & 7 & 4 \end{bmatrix}.$$

Answer / work



Question 14 (continued)

(c) Flatten all pooled channels into a list $x_1, \ldots, x_N$. Flattening only rearranges values. A fully connected layer has three outputs, each computed as $y_j = \sum_{i=1}^{N} w_{ji} x_i + b_j$. Find N and count the weights and biases in this layer. Count neither convolution parameters nor arithmetic operations.

Answer / work

Question 15: A supplied smoothing kernel

Apply the following kernel to the center of the given patch:

$$K = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}, \qquad I = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 10 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

Multiply corresponding kernel and patch entries and sum. You do not need to recall any named kernel from memory.

(a) Compute the output and explain the role of the kernel's sum. Would multiplying every kernel coefficient by 2 preserve a constant patch? For this part, consider a window entirely within the image, without padding.

Answer / work

Question 15 (continued)

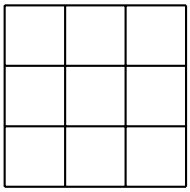
(b) Add one row or column of zeros on each side of I, giving the padded image

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 2 & 10 & 2 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Apply K with stride 1 and fill the entire 3×3 output grid. Show the top-left, top-middle, and center weighted sums. You may use symmetry for the other entries.

Answer / work

Filtered output

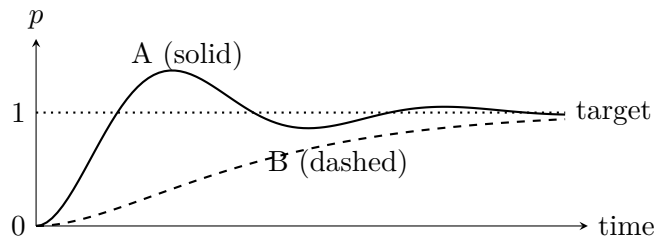


(c) Describe a useful effect and a possible disadvantage of smoothing before detecting an object. How does this fixed filter differ from filters learned in an AlexNet-style classifier?

Answer / work

Question 16: Interpreting PID and LQR responses

The plots show illustrative position responses of the same one-axis system to a target of 1, starting from rest at 0. The controllers use different gains. These curves are provided for interpretation, not for recovering numerical gains.



(a) Compare overshoot, oscillation, and approach to the target. For a PID controller with zero integral gain, propose one gain change that might reduce the oscillation in A. Explain a possible disadvantage and why a plot alone does not identify unique gain values.

Answer / work

Question 16 (continued)

(b) In a separate PID trial, a constant load opposes motion. Integral action is disabled, and the response settles below the target. Why can a position error remain after the motion stops? Explain what integral action could contribute and one undesirable effect to watch for when it is enabled. No controller equation is required.

Answer / work

(c) A supplied LQR solver chooses gains using weights on position error, velocity, and control effort. Compare two changes made separately from the same initial settings: increasing the position-error weight, and increasing the control-effort weight. Explain the intended tradeoff in each case and propose a comparison of the resulting responses. No objective function or gain calculation is required.

Answer / work

Question 17: A reading and an occupancy prior

A robot estimates whether one map cell is occupied. Let O denote occupied and F denote free. The prior is $P(O) = 0.2$, and the sensor has $P(\texttt{hit} \mid O) = 0.8$ and $P(\texttt{hit} \mid F) = 0.2$. Otherwise it reports `no hit`. Assume that the robot's pose is known, so the observed cell is unambiguous.

(a) The sensor reports `hit`. Use Bayes' rule to compute $P(O \mid \texttt{hit})$. Show the prior-times-likelihood products and their normalization. Explain why the answer is not 0.8.

Answer / work

(b) In a separate trial with the same original prior, the sensor reports `no hit`. Compute the posterior occupancy probability. State the likelihood for each hypothesis.

Answer / work

Question 18: Weighting is not resampling

Three predicted particles have prior weights $(1/2, 1/4, 1/4)$. The new observation has likelihoods $(1/5, 4/5, 2/5)$ at their locations. For systematic resampling, first form cumulative weights $\mathrm{cdf}_j = \sum_{k=1}^{j} w_k$. Particle j occupies the interval $[\mathrm{cdf}_{j-1}, \mathrm{cdf}_j)$, with $\mathrm{cdf}_0 = 0$. A draw selects the particle whose interval contains it. The random offset has already been chosen as $u_0 = 1/12$. Use draws u_0, $u_0 + 1/3$, and $u_0 + 2/3$. This procedure copies existing particle locations.

(a) Compute normalized posterior weights. Explain why normalizing the likelihoods alone gives a different answer.

Answer / work

(b) List the cumulative weights, draws, selected particles, and new weights. Select the first cumulative weight strictly greater than each draw. Does selecting each particle once show that the observation had no effect?

Answer / work

Question 19: Planning with an uncertain map and pose

A service robot has two plausible locations on opposite sides of a wall. Its estimator reports their mean as the current position. The map also contains a passage last observed before furniture was moved. The robot must choose its next action.

(a) A teammate proposes running A* from the mean and executing the resulting path. Explain two distinct assumptions that must be checked. Would changing A* to RRT resolve those issues by itself?

Answer / work

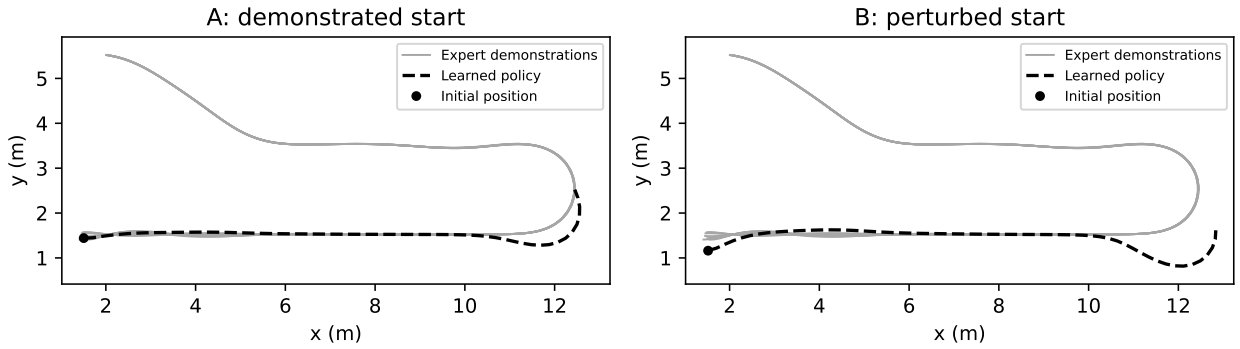
(b) Propose a defensible sequence of sensing and planning decisions. Explain where a reusable PRM or trajectory optimization might help, and what neither supplies automatically.

Answer / work

Question 20: Testing an imitation policy

A policy was trained to predict an expert driver's steering command from the car's pose. The figure shows the same policy driving from a demonstrated starting pose (A) and a displaced starting pose (B). After each motion, the policy receives the car's new pose and selects the next command. Both runs use the expert's speed.

Gray curves show expert demonstrations. Dashed curves show the learned policy, starting at the black dot. Both panels use the same scale.



(a) Compare the two trajectories. What does this experiment reveal that accurate steering predictions on held-out expert poses would not? Give one plausible explanation for the difference.

Answer / work

(b) Would collecting more successful expert laps necessarily address the problem? Propose one change to the training data and explain how you would test whether it improves recovery.

Answer / work

Question 21: Deployment safeguards ★ (required for 281; optional for 81)

A delivery robot uses a learned policy that sometimes leaves its marked route under unfamiliar lighting. No additional expert demonstrations can be collected. An independent range sensor and a pose estimate are available. A team compares three systems on the same 20 simulated routes, with the same starting conditions and lighting changes. Each trial ends at a collision, a completed delivery, or a fixed trial limit.

System	Collisions	Completed	Neither by trial limit
Original policy	8	12	0
Stop on a detected deviation	0	5	15
Switch to a feedback controller	1	17	2

(a) Recommend one system for further testing, stating your operating priorities. Explain what the results support and what they do not establish. Multiple recommendations may be defensible. In particular, does a zero in the collision column establish that the system has solved the navigation problem?

Answer / work

(b) Propose one modification that could improve the stopping system's ability to finish deliveries without collecting new expert demonstrations. State any additional information or capability it requires, one limitation, and a test that could reveal that limitation. Do not simply disable the stopping condition.

Answer / work